

PRIYANSHU UPADHYAY

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Education

KIET Group of Institutions

B.Tech in Computer Science and Engineering

Delhi NCR

2023 – Present

Kanha Makhan Public School

Senior Secondary (Class XII) — CBSE, 90%

Mathura, UP

2022 – 2023

Projects

Customer Retention Intelligence | FastAPI, XGBoost, Scikit-learn, PostgreSQL | [GitHub](#) | [Demo](#)

Mar – Apr 2026

- Engineered a thread-safe Scikit-learn pipeline with custom sequential preprocessors and a parallel ColumnTransformer (StandardScaler + OneHotEncoder) feeding XGBoost — achieving 85% recall and 0.84 ROC-AUC.
- Designed a Recall@20% budget-ranking strategy capturing ~50% of actual churners using only 20% of the retention budget, replacing naive threshold-based classification.
- Built a decoupled FastAPI inference engine with async PostgreSQL telemetry logging (BackgroundTasks + asyncpg), recording inference data without introducing UI latency.

QnA With Jupyter (Relational RAG System) | LangChain, ChromaDB, Groq, Ollama, Python AST | [GitHub](#)

Feb – Mar 2026

- Engineered a “Bait & Switch” retrieval pipeline: semantic search identifies a `cell_id`, then fetches the full uncut parent cell from a JSON ground-truth warehouse — eliminating context loss from naive chunking.
- Built dual-embedding ChromaDB pipelines — bge-m3 for NL semantics and CodeT5 for code syntax — and routed final Q&A through Groq API (Llama 3.3 70B) for low-latency inference; achieved ~80% contextual accuracy.
- Deployed local Llama 3.1 (Ollama) with Python AST-based ingestion to parse notebook dependencies and auto-generate semantic cell descriptions — keeping sensitive data off cloud APIs while enabling rich indexing.

LoanClassifier (Explainable AI) | XGBoost, SHAP, K-Means, LLM APIs | [GitHub](#) | [Demo](#)

Oct – Nov 2025

- Built a dual-model engine on 2.2M records (150 features): XGBoost for default probability prediction alongside K-Means clustering for dynamic financial risk segmentation.
- Integrated SHAP (Explainable AI) to quantify top features per loan decision; fed risk scores, cluster stats, and SHAP values into a context-aware LLM agent generating personalised financial advice and alternate loan terms.
- Rejected SMOTE after A/B testing revealed calibration noise in high-probability prediction bounds; used native XGBoost class weights instead, improving reliability on the imbalanced 2.2M-record dataset.

Technical Skills

Programming: Python, C++, C

ML & Data: Scikit-learn, XGBoost, Pandas, NumPy, SHAP (Explainable AI), Feature Engineering

Generative AI: LangChain, RAG Pipelines, LLM Integration, Prompt Engineering, ChromaDB (Vector DB)

Deployment & APIs: FastAPI, Streamlit, Docker, REST APIs

CS Fundamentals: OOP, OS, DBMS, SQL

Tools: Git, PostgreSQL, Jupyter Notebook, VS Code, OpenCV

Key Achievements

- Clash of Codes** — Finalist, Top 1% (1000+ participants, 256+ teams): built an AI virtual canvas enabling air-drawing via hand gesture recognition (OpenCV), with voice-controlled navigation, multi-color support, and Gemini integration to solve problems drawn on the canvas.
- HackSpire, IPEC Ghaziabad** — Finalist: developed a computer vision-based water footprint calculator that identifies crop types from images and estimates agricultural water consumption to guide sustainable irrigation decisions.

Position of Responsibility

Machine Learning Lead

May 2025 – Present

Technocrats KIET

Delhi NCR

- Managed a 5+ member team to plan and run ML/AI bootcamps for 100+ students, overseeing technical execution and troubleshooting across the full pipeline from data processing to model deployment.
- Mentored 5+ students on project architecture and model debugging, projects were related to end to end models, computer vision and GenAI.

Coordinator

Oct 2024 – May 2025

Technocrats KIET

Delhi NCR

- Organised ML workshops and a 3-day Computer Vision bootcamp attended by 100+ students.
- Taught foundational ML concepts through to model training, guiding students from basics like supervised learning and data preprocessing to building and training their first models.

Certifications

- AWS Certified AI Practitioner** [\[Verify\]](#)

Feb 2026

- Linear Algebra for Machine Learning and Data Science** — DeepLearning.AI [\[Verify\]](#)

Jan 2025